

Changyeon Jo

System Software Engineer
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INDUSTRY EXPERIENCES	MangoBoost Inc. , Seoul, South Korea	Nov 2022 - Present
	<i>System Software Engineer</i>	
	<i>Solution Evaluation Team, Lead</i>	2024 - Present
	Lead a team of 10 engineers performing system-level performance evaluation and validation of DPU products across diverse workload scenarios, with a primary focus on network performance and scalability. Led the company's MLPerf Storage benchmark submissions. In v2.0, Mango StorageBoost sustained line-rate throughput over a 400G Ethernet fabric and delivered 6.2x GPU scalability and 1.57-2.05x higher throughput per unit of network bandwidth over competing solutions, outperforming NVIDIA BlueField-3 in both NVMe/TCP and NVMe/RDMA configurations. In v1.0, the submission averaged 2.7x (up to 4.7x) higher performance than other submitters on ResNet50 and 3D-UNet.	
	<i>Software Team</i>	2022 - 2023
EDUCATION	Developed device drivers for the in-house RoCEv2 IP and performed system-level validation and testing of the IP in integrated environments.	
	Samsung Electronics , Hwaseong-si, South Korea	Sep 2021 - Oct 2022
	<i>Memory Software Development Team, Staff Engineer</i>	
	Built a cloud-based data pipeline to ingest, clean, and process globally collected real-time SSD diagnostic data, and developed failure-pattern analyses and failure prediction models over approximately ten years of accumulated diagnostics.	
	WorldQuant , Seoul, South Korea	Jun 2019 - Aug 2019
	<i>Research Consultant</i>	
	Developed stock price prediction models using WorldQuant's internal alpha research tools.	
	Seoul National University , Seoul, South Korea	
	<i>Ph.D. in Computer Science</i>	Mar 2012 - Aug 2021
	Thesis: <i>Remote Memory for Virtualized Environments</i> Advisor: Prof. Bernhard Egger	
RESEARCH EXPERIENCES	Hanyang University , Ansan, South Korea	
	<i>B.S. in Computer Science</i>	Mar 2008 - Feb 2012
	Capstone Project: <i>Musical Chords Generation for Given Melody using HMM</i>	
	ETH Systems Group , Zürich, Switzerland	Mar 2018 - May 2018
	<i>Visiting Ph.D. Student</i>	
PROJECTS	Project: Contributed to VMKit, the Systems Group's virtual machine monitor, adding modern x86 processor support. Advisor: Prof. Timothy Roscoe	
	PLASSE Lab , Hanyang University, Ansan, South Korea	Jul 2010 - Jun 2011
	<i>Undergraduate Intern</i>	
	Project: Survey on program analysis techniques. Advisor: Prof. Kyoung-Goo Doh	
	FORMULA: A Fast, Scalable and Flexible AI RNIC	
	An FPGA-based RDMA NIC for large-scale AI clusters, built around a line-rate RoCEv2 engine. The design adds multi-path RDMA transport and a programmable congestion control framework to sustain performance at scale, and a configurable block that offloads data-centric processing tasks at line rate, in a low-profile form factor with software compatibility for deployability. To appear at MICRO-59, 2026.	
	RoCEv2 Device Driver and Validation	2022 - 2023
	Developed the host device driver stack for an in-house RoCEv2 IP and built system-level validation for the IP in integrated environments.	

Instant Virtual Machine Live Migration 2018 - 2021
Remote memory gives a unique opportunity for virtual machine (VM) live migration by eliminating the memory copy required to move a VM to another machine. We propose an optimized VM live migration technique for remote memory environments. Our technique completes a VM migration in less than 100ms for memory-intensive workloads.

Remote Memory for Virtualized Environments 2018 - 2020
Using remote memory for efficient resource utilization is rapidly getting attention with the rising popularity of high-performance network. In this project, we proposed a tailored remote memory for virtualized environments. Our system reduced remote paging latency by 41.7x at the tail and improved job execution time by 3.5x under intensive remote paging scenarios.

Machine Learning Approach to Live Migration Modeling 2015 - 2017
VM live migration is the foundation of seamless management of cloud services. However, predicting its key performance metrics is notoriously difficult due to its complex behavior. In this project, we proposed a machine learning (ML) approach to live migration modeling. With the 40,000 VM live migration data, the ML model showed 2 to 5 times better prediction accuracy than the state-of-the-art analytical model.
Project page: <https://csap.snu.ac.kr/software/lmdataset>

VM Checkpoint, Restoration and Live Migration Techniques 2012 - 2015
VM state management is an essential feature for optimizing user experience in virtualized environments. In this project, we proposed fast and space efficient techniques for checkpoint, restoration, and live migration. The proposed techniques reduced VM management overheads by 30% on average in the evaluation with real applications.
Project page: <https://csap.snu.ac.kr/software/xencheckpointing>

PUBLICATIONS Kanghyun Choi, Bogyong Park, Gihwan Lee, Hanmin Kim, Nayeon Kim, Chanmyeong Kim, Hyeonseong Choi, Dongjoo Lee, Jaehyun Lee, Eunjin Baek, Changsu Kim, **Changyeon Jo**, Minhoo Chu, and Jangwoo Kim. “MangoBoost FORMULA: A Fast, Scalable and Flexible AI RNIC.” *In the 59th IEEE/ACM International Symposium on Microarchitecture (MICRO'26), Industry Track*, Athens, Greece, 2026. To appear.

Youngsu Cho, **Changyeon Jo**, Reza Entezari-Maleki, Jörn Altmann, and Bernhard Egger. “Towards Smarter Live Migration: Minimizing SLO Violations and Costs.” *In Future Generation Computer Systems, Volume 175, 108085*, 2026.

Heetaek Jeong, Wonsik Lee, Eunjin Baek, Changsu Kim, **Changyeon Jo**, Dongju Chae, Kanghyun Choi, Hamin Jang, Mohamed A. Elgammal, Sungmin Hong, Eriko Nurvitadhi, Dongup Kwon, and Jangwoo Kim. “MangoBoost Alice: Extremely Fast, Seamless, and Versatile FPGA-Accelerated DPU Solutions.” *In IEEE Micro, Volume 45, Issue 5, pp. 43 - 55*, 2025.

Younghyun Cho, Jiyeon Park, Florian Negele, **Changyeon Jo**, Thomas R. Gross, and Bernhard Egger. “Dopia: Online Parallelism Management for Integrated CPU/GPU Architectures.” *In 27th ACM SIGPLAN Symposium on Principles and Practice of Parallel Programming (PPoPP'22)*, April 2-6, 2022, Seoul, Republic of Korea.

Hyunik Kim, **Changyeon Jo**, and Bernhard Egger. “RapidSwap: A Hierarchical Far Memory.” *Presented at the 18th International Conference on the Economics of Grids, Clouds, Systems and Services (GECON'21)*, Virtual Event, September 2021. In Lecture Notes in Computer Science (LNCS), Volume 13072, December 2021.

Daon Park, Hyeonsoo Kim, **Changyeon Jo**, and Bernhard Egger. “Can VM Live Migration Improve Job Throughput? Evidence from a Real World Cluster Trace” *Presented at the 18th International Conference on the Economics of Grids, Clouds, Systems and Services (GECON'21)*, Virtual Event, September 2021. In Lecture Notes in Computer Science (LNCS), Volume 13072, December 2021.

Changyeon Jo, Hyunik Kim, Hexiang Geng, and Bernhard Egger. “RackMem: A Tailored Caching Layer for Rack Scale Computing.” *In Proceedings of the 29th Inter-*

Virtual Event, October 2020.

Changyeon Jo, Hyunik Kim, and Bernhard Egger. “Instant Virtual Machine Live Migration.” In *Proceedings of the 17th International Conference on the Economics of Grids Clouds, Systems and Services (GECON'20)*, Virtual Event, September 2020.

Yongsu Cho, **Changyeon Jo**, Hyunik Kim, and Bernhard Egger. “Towards Economical Live Migration in Data Centers.” In *Proceedings of the 17th International Conference on the Economics of Grids Clouds, Systems and Services (GECON'20)*, Virtual Event, September 2020.

Changyeon Jo, Yongsu Cho, and Bernhard Egger. “A Machine Learning Approach to Live Migration Modeling.” In *Proceedings of the 2017 ACM Symposium on Cloud Computing (SoCC'17)*, Santa Clara, USA, September 2017.

Changyeon Jo, Changmin Ahn, and Bernhard Egger. “A Machine Learning-based Approach to Live Migration Modeling.” Presented at the *4th International Workshop on Efficient Data Center Systems (EDCS'16) co-located with ISCA'16*, Seoul, Korea, June 2016.

Bernhard Egger, Eunbyung Park, Younghyun Cho, **Changyeon Jo**, and Jaejin Lee. “Efficient Checkpointing of Live Virtual Machines.” In *IEEE Transactions on Computers (TC)*, Volume 65, Issue 10, pp. 3041 - 3054, January 2016.

Bernhard Egger, Erik Gustafsson, **Changyeon Jo**, and Jeongseok Son. “Efficiently restoring virtual machines.” Presented at the *IFIP International Conference on Network and Parallel Computing (NPC'2013)*, Guiyang, China, September 2013, in *Springer International Journal of Parallel Programming (IJPP)*, Volume 43, Issue 3, June 2015.

Changyeon Jo and Bernhard Egger. “Optimizing Live Migration for Virtual Desktop Clouds.” In *Proceedings of the IEEE International Conference on Cloud Computing Technology and Science (CloudCom'2013)*, Bristol, UK, December 2013.

Changyeon Jo, Erik Gustafsson, Jeongseok Son, and Bernhard Egger. “Efficient live migration of virtual machines using shared storage.” In *Proceedings of the ACM SIGPLAN/SIGOPS International Conference on Virtual Execution Environments (VEE'13)*, Houston, USA, March 2013.

Seonghun Jeong , Youngchul Cho, Daeyong Shin, **Changyeon Jo**, Yenjo Han, Soojung Ryu, Jeongwook Kim, and Bernhard Egger. “Random Test Program Generation for Reconfigurable Architectures.” In *13th International Workshop on Microprocessor Test and Verification (MTV)*, Austin, USA, December 2012.

GRANTS

Young Researchers Exchange Program between Korea and Switzerland, Swiss State Secretariat for Education, Research and Innovation (SERI), 2018

ACM SIGMOD Travel Grants, ACM Symposium on Cloud Computing, 2017

TEACHING EXPERIENCES

M1522.000800 System Programming, Seoul National University, TA	2018 Fall
M1522.000800 System Programming, Seoul National University, TA	2017 Fall
4190.308 Computer Architecture, Seoul National University, TA	2017 Spring
4190.203 System Programming, Seoul National University, TA	2015 Fall
4190.308 Computer Architecture, Seoul National University, TA	2014 Fall
4190.203 System Programming, Seoul National University, TA	2013 Spring
4190.203 System Programming, Seoul National University, TA	2012 Fall
4190.203 System Programming, Seoul National University, TA	2012 Spring

PROFESSIONAL SERVICES

Artifact Evaluation Committee, *International Conference on Languages, Compilers, Tools and Theory of Embedded Systems (LCTES)*, 2019

External Reviewer, *IEEE Transactions on Cloud Computing*, 2015

SKILLS

RDMA (RoCEv2), DPU, SPDK, NVMe-oF, NVMe/TCP, Linux kernel, QEMU/KVM, Xen,

C, C++, Python, performance analysis and benchmarking (MI Perf)